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2018, Article / Letter to editor (Interdisciplinary Science Reviews, 43, 2, (2018), pp. 192-207)

Doi link to publisher: <https://doi.org/10.1080/03080188.2017.1380468>

Version of the following full text: Publisher's version

Downloaded from: <https://hdl.handle.net/2066/191407>

Download date: 2025-01-06

Note:

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Investigating consciousness in the sleep laboratory – an interdisciplinary perspective on lucid dreaming

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ABSTRACT

During dreaming, we experience a wake-like hallucinatory reality, however with restricted reflective abilities: in the face of a bizarre dream environment, we do not realize that we are actually dreaming. In contrast, during the rare phenomenon of lucid dreaming, the dreamer gains insight into the current state of mind while staying asleep. This metacognitive insight often enables the dreamer to control own dream actions and the course of the dream narrative. Lucid dreaming allows for radically new methodological and theoretical approaches and has led to new insights in diverse scientific disciplines beyond classical sleep and dream research, including neuroscience, psychotherapy, philosophy, art, and sports sciences. Here, we review past research and the current knowledge on lucid dreaming. We present insights into the scientific work in a sleep laboratory and describe how lucid dreams can be induced through methodologies from diverse academic backgrounds including psychology, electrical engineering and pharmacology.

ARTICLE HISTORY

Received 17 January 2017
Accepted 13 September 2017

KEYWORDS

Sleep; EEG; lucid dreaming;
cognition; metacognition;
consciousness

1. Dreaming – a brief introduction

Our conscious experience during sleep markedly differs from that of wakefulness: We hallucinate, engage in loose associations and confabulations, we are convinced that bizarre situations are real – or at least are not uncomfortable with their bizarreness –, we are disoriented regarding time, place and persons, we lack concentration, insight and willpower, we experience strong and varying emotions, and afterwards we quickly forget most of these highly unusual experiences (Hobson, Stickgold, and Pace-Schott 1998).

On the physiological level, each night we cycle through different *sleep stages*. Distinct features such as rapid eye movements (REMs), specific brain activation patterns such as sleep spindles and k-complexes, or muscle atonia allow the sleep researcher to classify periods of sleep into these stages. The two main categories REM sleep and Non-REM (NREM) sleep are labelled according to sleep periods with and without REMs, and can further be subdivided.

For many years, dreaming was considered to be phenomenon occurring during REM sleep only. Shortly after the discovery of different sleep stages in the 1950s (Aserinsky

and Kleitman 1953), sleep laboratory experiments showed that 74–80% of REM sleep awakenings resulted in vivid dream recall, compared to only 7–9% of NREM sleep awakenings (Dement and Kleitman 1957a, 1957b). A strong hint for the connection of REM sleep and dreaming was found in the 1960s, when experiments with cats revealed brain stem regions that control REM sleep atonia, i.e. the nearly complete paralysis of the skeletal muscles during this sleep stage, which is thought to be a protective mechanism preventing the dreamer from acting out dream movements. Cats with brain stem lesions showed a surprising behaviour when falling into REM sleep, apparently hunting non-existing mice or fighting non-existing enemies while being nearly unaffected by external stimuli (Jouvet and Delorme 1965). Naturally, this behaviour was interpreted as REM sleeping cats acting out their dreams. In the 1980s, a similar phenomenon was observed in humans as well: Patients suffering from the so-called *REM sleep behaviour disorder* (RBD) showed excessive movements during REM sleep, closely related to their dream contents (Schenck et al. 1986). This parasomnia bears apparent risks, as RBD patients acting out their dream have been reported in several cases to try diving from the bed into furniture or walls, choking or headlocking their bed partner or caregiver, or jumping out of a window (Schenck et al. 2009).

Already in the 1960s and 1970s, it became increasingly clear that dreaming is not a REM sleep exclusive phenomenon, but occurs in other sleep stages as well. The way how subjects are asked about their mental sleep experiences played an important role in this regard: Changing the question from ‘tell me if you had a dream’ to ‘tell me anything that was going through your mind just before you woke up’ resulted in much more frequent reports (up to 74%) of mental activity during NREM sleep (Foulkes 1962; Rechtschaffen 1973). However, NREM dreams are – especially in the first half of the night, when deep sleep is prevalent –, more thought-like, shorter, less visual, less motorically animated, less emotional and less vivid (Fosse, Stickgold, and Hobson 2001).

Even though the existence of NREM dreams is largely undisputed today, REM sleep dreams can be particularly well explained by their neurobiological basis. While some modern neurobiological dream theories support the traditional Freudian view of dreams as being the royal road to the unconscious, revealing unconscious desires related to egoistic and often infantile sexual impulses (Freud [1900] 1965; Solms 2000, 2004), other theories strictly doubt this. The highly influential activation-synthesis model (Hobson and McCarley 1977) and its revised form, the activation-input-modulation model (Hobson, Pace-Schott, and Stickgold 2000), consider dreaming not as a goal-directed messenger from the unconscious, but rather as the result of basic neurobiological activations devoid of deeper meaning: during REM sleep, the cerebral cortex is chaotically stimulated by brain stem regions via so-called Ponto Geniculate Occipital waves, which randomly activate cortically stored perceptions, memories, and motor sequences. The cerebral cortex as a self-organizing system then tries to make sense out of these random activations of cognitive elements, and synthesizes them into a relatively coherent narrative in the best possible way. Due to the chaotic input, the result is a surrealistic storyline full of bizarre elements and transitions.

Since the 1990s, neuroimaging techniques enabled new insights into the neurobiology of dreams and the conscious experiences therein (Nir and Tononi 2010; Dresler et al. 2014). For example, it was found, that visual association cortices and limbic regions including the amygdala are strongly activated during REM sleep, nicely explaining the

visual hallucinations and the strong emotions during dreaming. In contrast, cognitive deficits experienced in typical dreams such as impaired reflective thought and logical reasoning can be mapped to deactivations of the dorsolateral prefrontal and frontopolar cortices, which are involved in these functions during wakefulness (Hobson and Pace-Schott 2002).

2. Lucid dreaming

A special kind of dreaming differs from the impaired reflective capacity that characterizes normal dreams: During the so-called *lucid dreaming*, the dreamer realizes that he or she is dreaming while staying asleep (Van Eeden 1913). More sophisticated definitions of lucid dreaming demand that further criteria are fulfilled, such as full access to intellectual capabilities including waking life memories or the ability to exert control over the dream plot (Barrett 1992; Metzinger 2003; Windt and Metzinger 2007). Different questionnaires have been developed to assess such subtle degrees of dream lucidity (e.g. Kahan and Sullivan 2012; Voss et al. 2013). Lucid dreams thus frequently provide intentional influence over an internal virtual reality (Dresler et al. 2014), enabling the skilled lucid dreamer to freely pursue arbitrary actions in arbitrary environments with arbitrary people in arbitrary times. Accordingly, lucid dreams are used for a variety of aims. A survey among 301 lucid dreamers revealed that the most frequent application of lucid dreaming is just *having fun*: More than 80% of the lucid dreamers reported to have used their lucid dreams at least once for purposes such as flying, having sex, or other hedonistic activities: ‘One of the nicest things was that lemon cake which I enlarged to the size of a house. The best thing was that it really tasted like lemon cake!’ (Schädlich and Erlacher 2012). Besides *having fun*, lucid dreamers reported to use their lucid dreams for *changing a bad dream or nightmare into a pleasant one* (64%), *solving problems* (30%), *getting creative ideas or insights* (28%) and *practicing skills* (21%).

Generally, lucid dreaming is a rare phenomenon: Only about 23% of individuals experience them once or more per month, and only about every second person can remember at least one lucid dream at all from his or her own experience (Saunders et al. 2016). For comparison, non-lucid dreams are recalled about once a week (Schredl 2008). Lucid dreams are common amongst children and young adolescents, but seem to become less frequent after puberty, which has been associated with the maturation of prefrontal brain regions (Voss et al. 2012). There are no significant differences between sexes regarding the lucid dreaming frequency (Watson 2001). However, the general dream recall ability is correlated with lucid dream frequency (Schredl and Erlacher 2004), as well as with the occurrence of nightmares (Spadafora and Hunt 1990). Moreover, regular practice of meditation was also found to promote lucid dreaming (Gackenbach, Cranson, and Alexander 1986; Hunt 1991). Lastly, greater lucidity in dreams or more frequent lucid dreaming is linked to personality traits such as a higher internal locus of control, but not higher creativity (Blagrove and Tucker 1994), openness to experience (Schredl and Erlacher 2004), to well-being and vividness of visual imagery (Wolpin et al. 1992), to need for cognition (Blagrove and Hartnell 2000), to field-independence (Gackenbach et al. 1985), and to dissociation, daydreaming, sleep paralysis, and multi-sensory imagery (Denis and Poerio 2017).

From a physiological perspective, lucid dreaming has been associated with a higher activation of the autonomic nervous system – e. g. breathing frequency, heart rate, and

skin conductance – in contrast to non-lucid REM dreams (LaBerge, Levitan, and Dement 1986). Brain activity of lucid REM sleep as measured by electroencephalography (EEG) appears to differ from non-lucid REM sleep in the alpha (Tyson, Ogilvie, and Hunt 1984), beta (Holzinger, LaBerge, and Levitan 2006), and gamma bands (Voss et al. 2009). In particular the latter finding suggests a plausible neurophysiological explanation for the phenomenal characteristics of lucid dreams: The EEG of lucid REM sleep shows similarities to the EEG of the waking brain in the 40 Hz gamma band over the dorsolateral prefrontal cortex (DLPFC), which is typically deactivated during normal REM sleep. This type of brain activity can also be observed in the waking brain during tasks that require higher cognitive functions and conscious awareness (Uhlhaas et al. 2009). Increased dorsolateral prefrontal activations during lucid REM sleep have been confirmed by a combination of sleep EEG measurements and functional magnetic resonance imaging (fMRI, Dresler et al. 2012). In addition to the higher activation of the DLPFC, the fMRI recordings also revealed a higher activation of the so-called precuneus during lucid REM sleep. This region in the medial parietal cortex has been associated with self-directed cognition during wakefulness (Cavanna and Trimble 2006), which appears plausible given the definition of lucid dreaming as a state characterized by conscious reflection on one's current state of mind. Further insights into the neurobiological correlates of lucid dreaming might be gathered by network analyses of brain activity, as Spoormaker et al. suggest (2010a, 2010b).

Given the elusive character of dreams in general and the rarity of lucid dreams in particular, how can we be sure that lucid dreaming is a real phenomenon, and not based e.g. on false memories of entirely normal dream experiences? In the following, we will answer this question by describing the procedures in a typical night of lucid dreaming research in the sleep laboratory.

3. A night in the sleep laboratory

Lucid dreams are difficult to study due to their rarity – sleep researchers can hardly record weeks of sleep just to catch a single lucid dream. Consequently, in most lucid dreaming sleep laboratory studies participants are selected based on their lucid dream frequency, thereby increasing the chance to record a lucid dream in a given night. Subjects with frequent lucid dreams, which in addition have to be long enough to conduct scientific experiments within them, are hard to find. It is thus not uncommon to invite participants from the whole country or even from abroad for a lucid dreaming study, requiring effort for advertisement, recruitment, payment, and travelling.

Preparation and performance of actual lucid dreaming experiments in the sleep laboratory are complex as well – besides the usual extensive procedures necessary to conduct sleep laboratory studies, including as a minimum polysomnographic measurements of the brain activity by EEG, of eye movements by electrooculography, and of muscle activity by electromyography. For these measurements, electrodes are fixed on the scalp and in the face of the subject. The polysomnographic recording is necessary to determine whether a subject is awake or asleep, and also in which of the different sleep stages the subject is in, with standardized scoring rules being applied for sleep stage classification. Further, the research question in lucid dreaming projects often requires more detailed measurements of brain activity than is measured in normal sleep experiment settings, e.g. for elucidating

specific neurobiological correlates of a certain lucid dreaming task. As a result, preparation for the experiment can take several hours before the subject is sent to bed. In lucid dreaming studies with high-density EEG for topographical analyses or source localization of brain activity, for example, it takes two experimenters about three hours to instruct and prepare the subject, including fitting of the 128 EEG electrodes and measuring their positions and orientations.

Then the actual sleep recordings begin: In contrast to daytime EEG studies, where the subject usually sits still in front of a computer screen, sleeping subjects are turning around frequently, moving their heads, rubbing them into the pillow, pulling cables, and so on. Thus, lucid dreaming experimenters usually need to stay awake for the whole night in order to continuously supervise the quality of the recordings and to intervene in case of bad signal quality. There is nothing more frustrating for a lucid dream researcher than a bad EEG signal during the critical few minutes of a lucid dream. However, given that the preparation was successful and the subject finally experiences a lucid dream: How can this subjective experience be validated with objective scientific methods?

Even though the phenomenon of lucid dreaming has been described in the scientific literature already more than one century ago (Van Eeden 1913), it took more than 50 years until Green (1968) proposed a method for objective measurement of lucid dreams, which was ten years later put into practice by Hearne (1978) and LaBerge (1980a). According to the so-called scanning hypothesis (Ladd 1892), eye movements during REM sleep largely correspond to the subjectively experienced dream eye movements. Hearne and LaBerge asked their subjects to move their eyes in a predefined left-right-left-right (LRLR) manner as soon as they would become aware of their dreaming state, i.e. lucid. After awakening, a dream report was collected – and the previously agreed eye movements were consistently observed in both the dream report and the electrooculogram, clearly distinguishable from normal eye movements of REM sleep (LaBerge et al. 1981). These findings have been replicated in numerous studies since then, and allow for objective and exactly timed measurements of lucid dreams (Erlacher 2007; Dresler et al. 2011).

Having quickly developed into the gold standard in lucid dream research, also in our studies we use these LRLR eye signals to objectively verify that the subject actually achieved a lucid dream: eye movements are recorded and displayed on a computer screen in real-time, allowing for the experimenter to immediately identify a lucid dream whilst the subject is still asleep. After a predetermined time (or signs that the subject lost dream lucidity again), the successful lucid dreamer wakes up on his or her own, or is awakened by the experimenter in order to prompt a dream report and prevent forgetting of the dream. Besides writing down the dream report, also questionnaires about degrees of lucidity are filled out, and specific questions about the experiments conducted within the lucid dream are answered, before the subject goes back to sleep again for another lucid dreaming attempt.

The necessity to wake up subjects in case of a lucid dream without self-initiated awakening requires the experimenter to continuously monitor the eye recordings on the computer monitor throughout the whole night. While unfortunately still no automatized solutions for lucid dreaming LRLR eye signal detection in the sleep laboratory exist, some researchers have recently started developing such tools by applying methods from

the field of machine learning and pattern recognition to automatically detect lucidity-indicating eye movements or bad EEG signals in real-time (e.g. Appel 2013).

In the morning, when the subject declares that he or she cannot sleep any more, recordings are stopped, and it usually takes another one or two hours to let the subject fill out more questionnaires, to tidy up and clean the equipment, or to take a few other measures depending on the experiment. For example, a classical study protocol for motor skill training during lucid dreaming needs performance measurements of the chosen skill both in the evening and in the morning. All in all, the whole experimental procedure of recording one night including preparations and follow-up usually takes from 8 pm until 11 am the next day – for a few recording minutes of data of interest in the best case scenario, i.e. the subject experiencing an actual lucid dream. In many cases, two or three consecutive nights are recorded in a row in order to reduce travelling demands on the subject, resulting in an extensive workload for the lucid dream researcher. Given all these efforts and constraints, it is not surprising that lucid dreaming studies often suffer from a limited number of participants – if exceeding the character of case-studies of single subjects at all.

4. The variety of lucid dreaming induction methodologies

As our illustration of a typical night in the sleep laboratory demonstrates, recording a lucid dream is an effortful procedure – if succeeding at all, as an entirely non-lucid night is even in experienced lucid dreamers the more usual case. Development of a reliable method allowing to induce lucid dreaming on demand would thus imply strong progress for the field of lucid dreaming research. However, even though during the past 40 years numerous studies tested methods from different disciplines such as psychology, electrical engineering or pharmacology in search for this ‘Holy Grail’ of lucid dreaming research, no reliable method has been found yet (for an extensive overview see Stumbrys et al. 2012). Lucid dreaming is considered to be a learnable skill, even though it can take from days to months until the first lucid dream is experienced by the novice (LaBerge 1980b; Tholey 1983). Most lucid dream induction methods can be used stand-alone or in combination with each other. In the following we will discuss exemplary methods in order to demonstrate their variability, different methodological approaches and multidisciplinary backgrounds.

Psychological methods

Psychological methods for inducing lucid dreaming have been investigated most frequently (Stumbrys et al. 2012). These methods include all cognitive and behavioural activities that are carried out to increase the likelihood of lucid insight into the dream state. *Tholey’s combined technique* (Tholey 1983), for example, makes use of intention (imagining being in a dream and recognizing this), autosuggestion (repeatedly suggesting oneself to become lucid when falling asleep), and reflection, i.e. the development of a critical-reflective frame of mind concerning one’s state of consciousness: by repeatedly asking oneself throughout the day whether one is dreaming, a habit of critical self-awareness is developed. After some time, this self-reflexive habit will be transferred to the dream state and might thus prompt lucid insight into this state. However, on a generally uncritical background cognition that prevails during most dreams, such reflection on the

current state of mind does not necessarily lead to actual dream lucidity. In order to systematically test whether one is currently in a dream or awake, therefore a number of so-called *reality checks* have been developed. One classic example is to read a short text within a dream, look away from the text and back again – the text often changes its content during this procedure (LaBerge and DeGracia 2000). A further reality check would be to hold one's nose and try to breathe through it nevertheless, which obviously does not work while being awake, but is possible surprisingly often during a dream (this might be the case because the sleeping body is in fact breathing through the nose). *Tholey's combined technique* has been found to successfully increase the probability of lucid dreams: A field study by Paulsson and Parker (2006) with 20 participants compared a baseline week with no application of the technique to two weeks of using *Tholey's combined technique*. During the baseline week, on average only about 2% of the nights included a lucid dream. During the first week of applying the induction technique, 11 participants had lucid dreams (5 of which never had a lucid dream before), summing up almost one lucid dream per week. The number of lucid dreams further increased during the second week, resulting in also statistically significant increase in lucid dreaming frequency compared to the start of the study.

Numerous other cognitive techniques exist, for example the mnemonic-induced lucid dream (MILD) technique (LaBerge 1980b), which was tested most often empirically so far in scientific experiments. MILD requires the dreamer to rehearse a dream before falling asleep and to visualize achieving lucidity, combined with autosuggestion. A different cognitive approach is chosen in the so-called wake-induced lucid dreaming techniques, in which – in contrast to the previous described methods – the dreamer does not try to gain insight into the dreaming state during the dream, but tries to maintain this awareness already when falling asleep, e.g. by concentrating on the hypnagogic imagery at sleep onset (LaBerge and Rheingold 1990).

Pharmacological methods

Besides psychological induction methods such as *Tholey's combined technique*, also pharmacological approaches have been developed to directly alter the brain's chemistry such that lucid dreams become more probable. In one of the very few studies until now, LaBerge (2004) administered Donepezil, i.e. a drug from the class of acetylcholine esterase inhibitors usually indicated for patients suffering from Alzheimer's disease. This drug increases the level of the neurotransmitter acetylcholine in the body and in the brain, and thereby also affects lucid dreaming, as demonstrated in a double-blind placebo controlled experiment: In the Donepezil conditions (5 mg or 10 mg), nine out of ten subjects reported one or more lucid dreams during two nights, whereas only one out of ten subjects reported a lucid dream in the placebo night. As with most prescription drugs, however, Donepezil comes with side effects – for example more than one out of ten patients will experience nausea, other side effects include heart rate changes, cramps, hallucinations, or insomnia. Thus, despite its promising lucid dream induction capabilities, Donepezil is not well suited for lucid dreaming studies or private use for lucid dreaming purposes at home due to its side effects. Alpha-Glycerolphosphorylcholine, a non-prescription alternative to Donepezil that is thought to act on the acetylcholine system with less side effects, was recently tested with no significant results (Kern et al. 2017).

This shows that even though pharmacological strategies are promising, more research is needed in order to find a both reliable and safe drug for lucid dream induction.

Technical devices

During the past 40 years of lucid dream research also technical approaches for lucid dream induction have been tested, frequently leading to the development of commercially available consumer products. Most prominent in this regard are sleep masks that emit light effects or auditory cues during REM sleep. The idea behind this strategy is that these signals get incorporated into the dream and remind the dreamer of his or her plan to become lucid. The main difficulty thereby lies in the correct adjustment of the stimulus intensity: if the stimulus is too bright or loud, the subject wakes up, if it is not intense enough, it does not affect the dream. Studies focusing on light stimulation during REM sleep alone without combining it with e.g. cognitive methods showed only poor results of around 5% success rate per night (LaBerge 1988; Paul, Schädlich, and Erlacher 2014). Other stimulation types such as vibro-tactile (Reis 1989), electro-tactile (Hearne 1983), vestibular (Leslie and Ogilvie 1996) or water splashes in the face (Hearne 1978) have been investigated as well, however did not lead to reliable solutions for lucid dream induction on demand.

Electrical brain stimulation

Instead of using lights or sounds to indirectly alter dreaming, recent studies used transcranial direct current stimulation (tDCS, Stumbrys, Erlacher, and Schredl 2013) or transcranial alternating current stimulation (tACS, Voss et al. 2014) for lucid dream induction. For such electrical brain stimulation methods, electrodes are attached to the scalp and a small electrical current below the perception threshold is applied. The aim is to evoke neuronal activity in cortical areas which are associated with lucid dreaming, e.g. the DLPFC (see above). Since the subject is asleep and side effects can occur, this technique is not (yet) suited for home use. The results of the two studies conducted about this new lucid dream induction method look promising: compared to sham (placebo) stimulation, both tACS and tDCS slightly increased dream lucidity as measured with a dream questionnaire or via the eye signalling technique. However, the effect sizes especially in non-frequent lucid dreamers were rather low, and further improvements of this methodology are needed.

Unfortunately, none of the psychological, pharmacological or technical induction techniques developed so far induces lucid dreams both reliably and safely on demand, rendering lucid dreaming experiments in the sleep laboratory an effortful and time-consuming endeavour.

5. Lucid dreaming – an interdisciplinary field of research

Lucid dreaming is not restricted to private use, but has been studied as both a scientific tool and topic in different academic disciplines. In the following we will briefly overview how lucid dreaming research has been conducted in different fields such as neuroscience and psychology, sport science, arts, therapy, and philosophy.

Neuroscience and lucid dreaming

One example of a neuroscientific research problem, which was solved using lucid dreaming experiments, is the question whether the neuronal activity of the brain during waking experiences is similar to that of analogous, but merely dreamed experiences. The main obstacle to answer this question lies in the elusiveness of dreams: every dream is different, sleeping subjects normally cannot be asked to dream a specific content, and researchers do not know when exactly which element of the dream report happened in the respective sleep recording. With lucid dreaming, these problems can be solved: experienced lucid dreamers were asked to perform particular hand movements during their lucid dreams, and again later during wakefulness. Lucid eye signals were used as exact temporal markers of the tasks executed during dreaming. Subsequent analysis of the recorded brain activity could thus be performed exactly time-logged to the dreamed task performance, demonstrating that brain processes are indeed very similar between dreaming and wakefulness (Erlacher, Schredl, and LaBerge 2003; Dresler et al. 2011). Another neuroscientific experiment aimed to answer the old question whether the flow of time experienced during dreaming corresponds to objective time during wakefulness. To test this question, again lucid dreamers were instructed to use intentional eye movements during the dream as markers for the start and end of dream actions. By doing so, it could be demonstrated that the time needed for counting in a lucid dream is comparable to the time needed for counting in the waking state (LaBerge 1985). Interestingly, motor activities such as performing squats, walking, or a gymnastics routine appear to require slightly more time during dreaming compared to wakefulness, which might be due to the absence of muscular feedback or slower neural processing during REM sleep (Erlacher & Schredl, 2004; Erlacher et al. 2013).

Therapy and lucid dreaming

Viewed from a medical perspective, the *prima facie* potential of lucid dreaming as a therapeutic approach for nightmares is apparent: if the dreamer realizes during a nightmare, that the experienced threats are merely dreamed and thus not real, the nightmare becomes much less frightening. Moreover, the lucid dreamer can try to alter the dream story, letting the threats disappear, or change into something more pleasant. Various studies have examined this psychotherapeutic potential of lucid dreaming for nightmares, and also for post-traumatic stress, and supported its efficacy (Spoormaker, van den Bout, and Meijer 2003; Spoormaker and Van Den Bout 2006; Soffer-Dudek, Wertheim, and Shahar 2011; Holzinger, Klösch, and Saletu 2015). Besides using lucid dreaming for nightmare therapy, it might also help by serving as a new treatment approach for patients with schizophrenia. Dreaming has already been discussed as a natural model for psychosis for several centuries (Hobson 2004), and recent empirical work supports this idea (D'Agostino, Limosani, and Scarone 2012). One major symptom of psychotic patients is that they are not able to recognize their pathologic state as such – just as healthy subjects during dreaming typically do not recognize their current state of mind. In the dreaming-as-psychosis model, lucid dreaming thus represents the successful insight of the patient into his or her state. And in fact, psychotic patients with impaired insight into their pathologic state show anatomical and functional changes in exactly those brain

regions that are more active in lucid compared to non-lucid dreams (Dresler et al. 2015). On this background, lucid dreaming training might serve as a therapeutic approach to increase state insight also in patients with schizophrenia, and new schizophrenia drugs might be tested in healthy subjects with lucid dreaming frequency as a new outcome measure of interest.

Philosophy and lucid dreaming

Lucid dreaming has gained increasing interest from philosophers in recent years due to its implications for philosophical theories on consciousness. Common distinctions of this term differentiate between basal (primary) and higher (secondary) consciousness: Primary consciousness includes mostly sensations and perceptions, whilst secondary consciousness constitutes by reflecting about these basal contents (Edelman 2003). Insofar, gaining lucid insight into the dream state represents a very pure form of secondary consciousness, as the transition from primary to secondary consciousness happens within a comparable physiological milieu (Dresler et al. 2009; Hobson 2009; Hobson and Voss 2010). Further, the phenomenon of lucid dreaming is seen as of great value particularly for philosophical theories on subjectivity and the self, as lucid dreams represent a unique class of phenomenal states, namely the full experience of the current subjective state as being a merely simulated model of the world (Metzinger 2003, 2009; Windt and Metzinger 2007).

Sports science and lucid dreaming

Dreams in general can be seen as a perfectly simulated virtual reality (Revonsuo 1995). This makes lucid dreams, in turn, highly interesting for sport sciences, since they offer exceptional training opportunities (Erlacher and Chapin 2010). In particular extreme sports with high risks of injury benefit from the safe training environments that allows a risk-free rehearsal of complex movements. Few studies have tested the potential of lucid dreams for sports training, however with promising results. For example, it could be shown that it is possible to practice movements during a lucid dream and thereby improve also the waking performance of this task (Erlacher and Chapin 2010; Erlacher and Schredl 2010; Erlacher 2012; Schädlich, Erlacher, and Schredl 2017; Stumbrys, Erlacher, and Schredl 2016). Moreover, a survey among 840 German athletes revealed that 25% of them experience lucid dreams regularly, and that around 10% of the athletes use lucid dreaming for training purposes and to improve their performance during wakefulness (Erlacher, Stumbrys, and Schredl 2012). It thus appears that the promising sports training options of lucid dreaming are already used by a significant number of athletes.

Arts, music and lucid dreaming

Lucid dreaming is valued by artists because of its potential for incubating or boosting creative processes (Barrett 2001). Dreams in general are reported to have been the source of many artistic works or musical pieces – not only in surrealism, which explicitly uses dreams as a theoretical basis, but also in many other artistic directions (for an overview see Dresler 2008). Systematic studies about incubative processes of creativity confirm

these anecdotal evidences (Marisch et al. 2016). One reason for this lies in the neurophysiological milieu that harbours dreaming, and that provides many prerequisites considered to be important by psychological and neurocognitive theories of creativity: a variable cortical arousal, unusual connections between distant brain regions, flat associative hierarchies and a state of defocused attention (Dresler 2012; Marisch et al. 2016). Lucid dreaming offers a particularly creativity-enabling state: like in non-lucid dreams, the dreamer experiences a hyper-associative state with unusual combinations of ideas and sensations, however in contrast to a non-lucid dream he or she is capable of deliberately evaluating these new impressions with respect to their artistic value. And in fact, a number of painters and musicians use their lucid dreaming abilities for inspirational purposes – e.g. an artist looking for new scenes in a virtual dream gallery (Barrett 2001) or a musician looking for new melodies in lucidly dreamed radio programmes (Dobe 2012).

6. Conclusions and future directions

Lucid dreaming is both topic and tool in various scientific disciplines, from neuroscience to philosophy to sports science to arts to therapy. However, further research on the induction of lucid dreams, in the sleep laboratory and at home, is necessary. Despite numerous tested approaches from various academic backgrounds including psychology, electrical engineering or pharmacology, until now no method has been found to induce lucid dreams reliably and safely on demand for everybody and in every desired night. Besides the development of completely new ideas for lucid dream induction, work in this direction should also focus on further developing existing approaches. Combining and integrating approaches from different scientific disciplines might be the key to finding a reliable method for lucid dream induction – for example, by using supplements, technical devices and cognitive induction techniques simultaneously and by studying, which combinations of which methods increase the probability of lucid dreams best. It might be the case, however, that some induction techniques even reduce the effectiveness of other techniques, when used at the same time. From these interactions it might be possible to understand better, in which way each induction technique influences the lucid dream probability, why some methods work better for some people than for others, and whether a smart combination of techniques exists that induces lucid dreams in everybody.

Furthermore, more research should be conducted to investigate, how lucid dreams can be used most effectively for the various purposes described in this paper. This also includes research on ‘best practices’ of how lucid dreamers can influence their dreams best and how to transfer these best practices from one discipline to another. For example, if artists develop a good and reliable method for boosting and using creativity in lucid dreams, the same practice might be used for finding solutions to challenging problems in other applications.

Finally, a more advanced information transfer than simple LRLR eye signals from the lucid dream to the waking world is a promising aim for future research. With a bidirectional message exchange system (Appel 2013), ideas like a direct interaction between lucid dreams and the waking world for creating art, direct psychotherapeutic coaching during nightmares, or advanced neuroscientific sleep studies seem realizable in the nearer future.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by a grant from the Andrea von Braun Foundation.

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